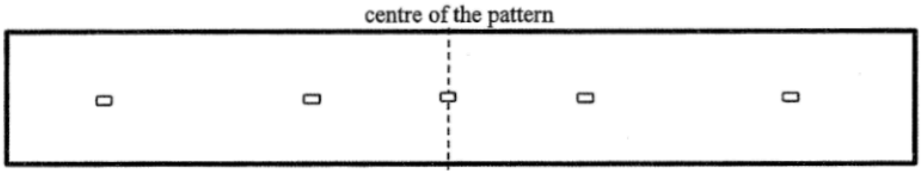
6. (a) (i)	$\Delta y = \dfrac{\lambda D}{a}$ $\dfrac{(4.0-0)\times 10^{-2}}{10} = \dfrac{\lambda(1.8)}{0.3\times 10^{-3}}$ $\lambda = 6.666667 \times 10^{-7}$ m $\approx 6.67 \times 10^{-7}$ m or 667 nm	1M+1M 1A 3	
(ii)	To ensure that the light through the 2 slits diffracts enough to interfere / overlap.	1A 1A 2	Explanation in terms of formula $\lambda = \Delta y \dfrac{a}{D}$ is **NOT** accepted as a is the slit separation given, which is fixed.
(b) (i)	$d \sin\theta = m\lambda$ $\dfrac{10^{-3}}{500} \sin\theta = 6.67 \times 10^{-7}$ m $\theta = 19.471221° \approx 19.47°$ Separation between central and first-order bright spot $= 1.8 \tan 19.47°$ $= 0.636396$ m ≈ 0.636 m	1M 1M 1A 3	

(ii)

centre of the pattern



	symmetry about the central bright spot (2nd order shown) separation between 2nd and 1st -order bright spots is larger	1A 1A 2	